

Configure AI large model

Configure AI large model

1. Overseas Version of OpenRouter
OpenRouter Operation Steps
2. Overseas Version of Tongyi Qianwen
3. Local DIFY Configuration
 - 3.1 Enter the API key.

For large-scale model examples, an important prerequisite is connecting to a Wi-Fi network with internet access. For details on how to connect, see "Raspberry Pi Basic Settings" under "Raspberry Pi System Configuration" - 04. Network Settings for specific instructions.

This tutorial is about configuring large AI models. To run large model-related examples properly, follow the steps below.

1. Overseas Version of OpenRouter

Some online large models in the overseas version startup examples use the middleware of the OpenRouter platform, a third-party platform. This large model can be used 50 times per day for free. If you wish to continue using it, refer to the usage rules for expansion.

The following examples require this large model: scene description, dual-model intelligent agent application, intelligent action choreography, and free conversation.

OpenRouter does not have free models for video analysis or text-to-image creation, so models from the Tongyi Qianwen platform are required.

OpenRouter Operation Steps

1. Log in to the official website:

<https://openrouter.ai/>

2. Log in. You can use a variety of methods, but this article recommends using a GitHub account.

The Unified Interface For LLMs

Better **prices**, better **uptime**, no subscription.

Start a message...



Featured Models

[View Trending](#)

Gemini 2.5 Pro Preview

New

by google

83.1B

Tokens/wk

13.9s

Latency

+7.42%

Weekly growth

GPT-4.1

New

by openai

39.9B

Tokens/wk

584ms

Latency

+13.06%

Weekly growth

Claude 3.7 Sonnet

by anthropic

324.5B

Tokens/wk

1.5s

Latency

-3.37%

Weekly growth

7.9T

Monthly Tokens

1.9M

Global Users

50+

Active Providers

300+

Models

3. Go to the API-KEY page and create one.

Credits

Keys

Activity

Settings

Sign out

The Unified Interface For LLMs

Better **prices**, better **uptime**, no subscription.

Start a message...



Featured Models

[View Trending](#)

Gemini 2.5 Pro Preview

New

by google

83.1B

Tokens/wk

13.9s

Latency

+7.42%

Weekly growth

GPT-4.1

New

by openai

39.9B

Tokens/wk

584ms

Latency

+13.06%

Weekly growth

Claude 3.7 Sonnet

by anthropic

324.5B

Tokens/wk

1.5s

Latency

-3.37%

Weekly growth

7.9T

Monthly Tokens

1.9M

Global Users

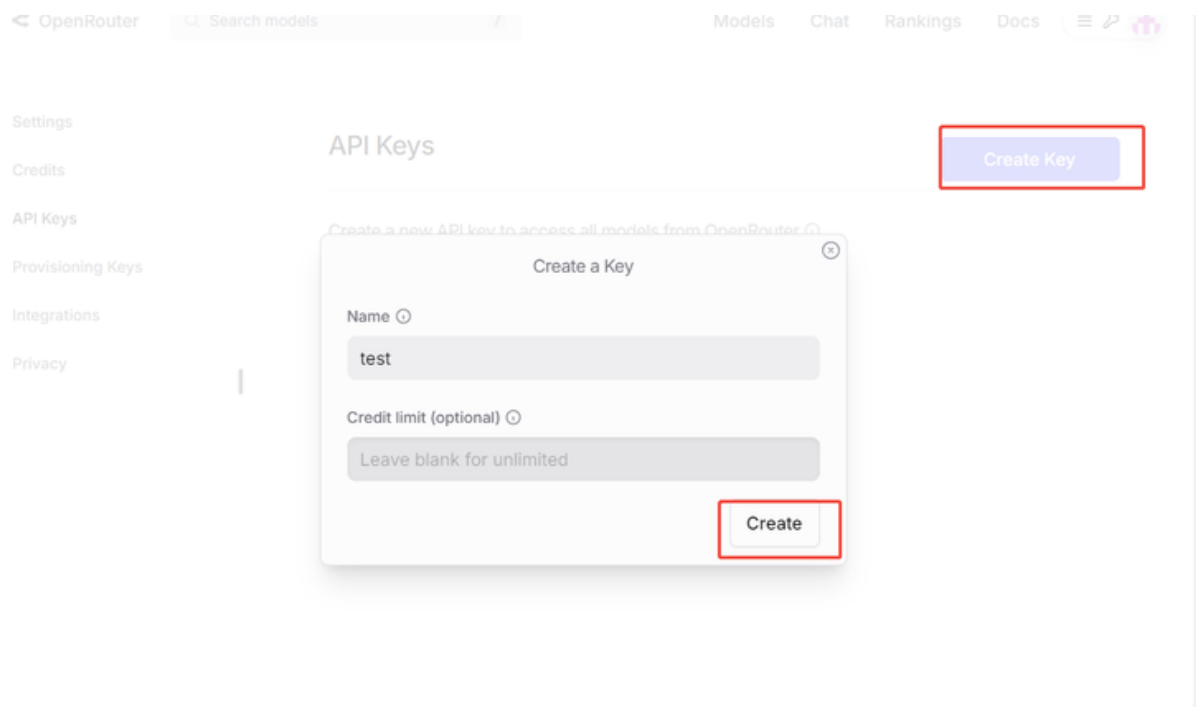
50+

Active Providers

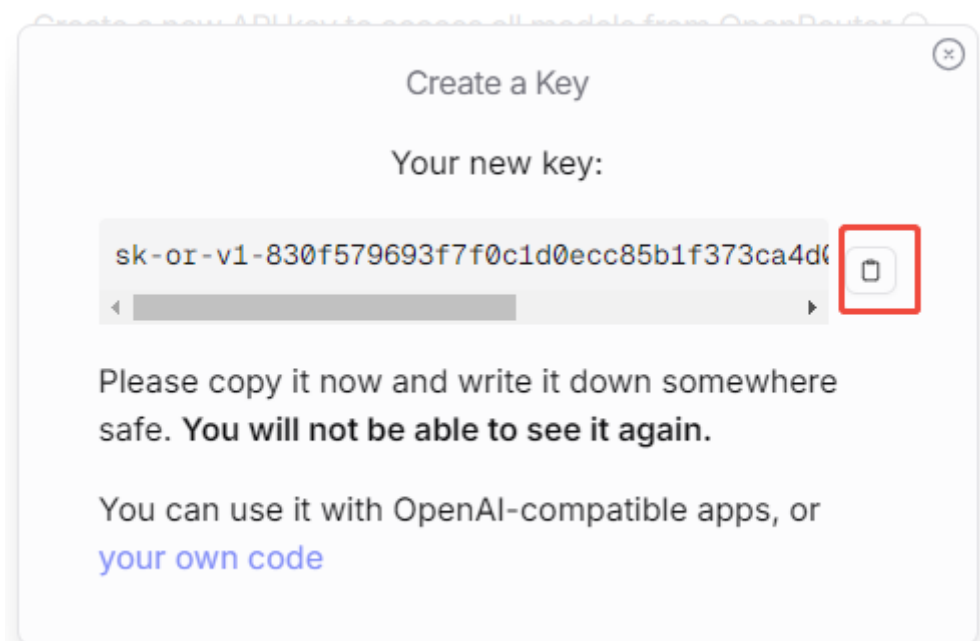
300+

Models

<https://openrouter.ai/settings/keys>



4. Copy and save the created key to your computer. **You can only copy this key the first time you create it;** you won't be able to view it again after leaving this page.



5. Next, we'll use a free test model to see if the registered account is functioning properly.

OpenRouter

Search models

/

Models

Chat

Rankings

Docs



The Unified Interface For LLMs

Better **prices**, better **uptime**, no subscription.

Start a message... 

Featured Models

[View Trending](#)

Gemini 2.5 Pro Preview

by google

83.1B Tokens/wk

13.9s Latency

+7.42% Weekly growth

GPT-4.1

by openai

39.9B Tokens/wk

584ms Latency

+13.06% Weekly growth

Claude 3.7 Sonnet

by anthropic

324.5B Tokens/wk

1.5s Latency

-3.37% Weekly growth

7.9T

Monthly Tokens

1.9M

Global Users

50+

Active Providers

300+

Models

OpenRouter

Search models

/

Models

Chat

Rankings

Docs



Input Modalities

Text

Image

File

Context length

4K

64K

1M

Prompt pricing

FREE

\$0.5

\$10+

Series

GPT

Claude

Gemini

Models

59 models

Reset Filters

free

Sort



DeepSeek: DeepSeek V3 0324 (free)



98.8B tokens

DeepSeek V3, a 685B-parameter, mixture-of-experts model, is the latest iteration of the flagship chat model family from the DeepSeek team. It succeeds the DeepSeek V3 model and performs ...

by [deepseek](#) | 164K context | \$0/M input tokens | \$0/M output tokens

DeepSeek: R1 (free)



44.2B tokens

DeepSeek R1 is here: Performance on par with OpenAI o1, but open-sourced and with fully open reasoning tokens. It's 671B parameters in size, with 37B active in an inference pass. Fully open...

by [deepseek](#) | 164K context | \$0/M input tokens | \$0/M output tokens

Google: Gemini 2.0 Flash Experimental (free)



31.2B tokens

Gemini Flash 2.0 offers a significantly faster time to first token (TTFT) compared to Gemini Flash 1.5, while maintaining quality on par with larger models like Gemini Pro 1.5. It introduces notable ...

by [google](#)

正在等待 openrouter.ai 的响应...

1.5

6. If the response is normal, it means it's functioning properly.

OpenRouter Search models /

Models Chat Rankings Docs

DeepSeek: DeepSeek V3 0324 (free)

deepseek/deepseek-chat-v3-0324:free

Created Mar 24, 2025 | 163,840 context | \$0/M input tokens | \$0/M output tokens

DeepSeek V3, a 685B-parameter, mixture-of-experts model, is the latest iteration of the flagship chat model family from the DeepSeek team. It succeeds the [DeepSeek V3](#) model and performs really well on a variety of tasks.

Free Model weights

Overview Providers Versions Apps Activity Uptime API

Providers for DeepSeek V3 0324 (free)

OpenRouter routes requests to the best providers that are able to handle your prompt size and parameters, with fallbacks to maximize uptime.

Sort by

Chutes	Context	Max Output	Input	Output	Latency	Throughput	Uptime
fp8	164K	164K	\$0	\$0	1.60s	40.37t/s	III

正在连接...

New Room To: DeepSeek V3 0324 (free) + Add model

hello,who are you?

hello,who are you?

DeepSeek V3 0324 (free) | Targon

Hello! I'm DeepSeek Chat, an AI assistant created by DeepSeek. My purpose is to help you with a wide range of tasks—whether it's answering questions, brainstorming ideas, assisting with coding, or just having a friendly chat. 😊

How can I help you today?

Web Search New Room

Start a message...

- The API key you create directly supports free models. If you wish to use non-free models, you will need to purchase the required amount with your credit card. This tutorial will not be explained in detail, so please research it on your own.

2. Overseas Version of Tongyi Qianwen

Some of the online large models in the overseas version startup examples use Tongyi Qianwen's large visual model. This platform is a third-party platform, and the large model is a powerful visual model provided by Alibaba. After registration and authentication, you can use some of the large model for free, or pay for additional model usage according to the website.

The following examples require this large model: dual-model intelligent agent-object tracking, video description, and text-based image creation.

1. Open the website and register.

Register account

The screenshot shows the Alibaba Cloud registration page. On the left, under 'Account Benefits', there are sections for 'Free Trial' (Get free hands-on experience with 50+ products. Now up to 12 Months for Elastic Compute Service!) and 'Premium Support Services' (1-on-1 pre-sale consultation, 24/7 after-sales technical support with 6 free tickets per quarter). On the right, the 'Sign up to Alibaba Cloud' form is displayed. It asks to 'Please select your account type *'. Two options are shown: 'Business Account' (For purchasing services required by businesses. Enjoy premium support services and exclusive offers.) and 'Individual Account' (For purchasing services required by individuals or for personal use.). The 'Individual Account' option is selected and highlighted with a red box. Below the options is a 'Next' button. There are also links for 'Sign up with Google' and 'Sign up with Github'. At the bottom, it says 'Already a member? Sign In'.

The screenshot shows the next step of the Alibaba Cloud registration process. The 'Sign up to Alibaba Cloud' form now has fields for 'Email Address *' (Enter your email), 'Password *' (Enter your password), and 'Confirm Password *' (Confirm your password). There are 'Sign Up (Step 1 of 2)' and 'Go Back' buttons. At the bottom, it says 'Already a member? Sign In'.

2. After registration, enter your contact information based on your country and verify your identity. Without verification, you will not be able to receive the free large model.

Alibaba Cloud

AI SearchContact SalesEnglishCartConsole

Why UsPricingProductsSolutionsMarketplaceDevelopersPartnersDocumentationServicesModel StudioComplete Sign Up

Successfully registered!

Keep update information.

Complete your account information to use Alibaba Cloud cloud computing services. You can complete the information later.
[Return to Previous Page \[+\]](#)

Security Verification

For account security reasons, please bind your current mobile phone number and use it only in security scenarios such as one time password (OTP) and password retrieval scenarios. [\[+\]](#)

Mobile Phone Number *

+64

Enter a mobile phone number

Verification Code *

Enter a verification code

Obtain Verification Code

Failed to obtain a verification code? [\[+\]](#)

Country/Region

New Zealand

☐ Alibaba Cloud may use your information to contact you about updates and special offers. You can unsubscribe at any time.

Next

Payment Information

3. Go to Large Models, select Qwen-VL-Max, and receive your free model. Withdraw some large model credits

Alibaba Cloud

AI SearchContact SalesEnglishCartConsole

Why UsPricingProductsSolutionsMarketplaceDevelopersPartnersDocumentationServicesModel StudioFree Trial

TiDB Cloud Available for Singapore region now

Creating the cutting-edge TiDB cloud-native architecture on Alibaba Cloud

Learn More

01 ApsaraDB for MongoDB 8.0 Released

02 Qwen3 Models

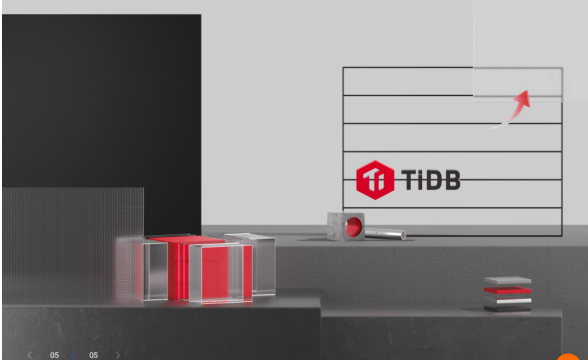
03 Simple Application Server Special Offer

04 Cloud Drive for Enterprises

Free Tier

Promo Center

Alibaba Cloud Model Studio



SAS - the New VPS ChoiceOnly \$9.9/yearSAS 2vCPUx 1GBBuy Now

Alibaba Cloud

AI SearchContact SalesEnglishCartConsole

Why UsPricingProductsSolutionsMarketplaceDevelopersPartnersDocumentationServicesModel StudioComplete Sign Up

Alibaba Cloud > Products > Alibaba Cloud Model Studio

Model Studio: Supercharge Your AI Journey Effortlessly With Industry-Leading GenAI Models

You can seamlessly integrate our advanced AI models—such as Qwen-Max (MoE), Qwen-VL (visual understanding), and the *Wan* series (video generation)—via Model Studio's purpose-built APIs

Activate Now

Contact Sales

Qwen2.5 and Qwen2.5-VL Open-Source Free Trial Models will be commercialized from 2025-06-04 00:00:00 (UTC+08). Click here for more details.

Introducing Qwen3

Why Us

Features

Scenarios

Models and Pricing

Customer

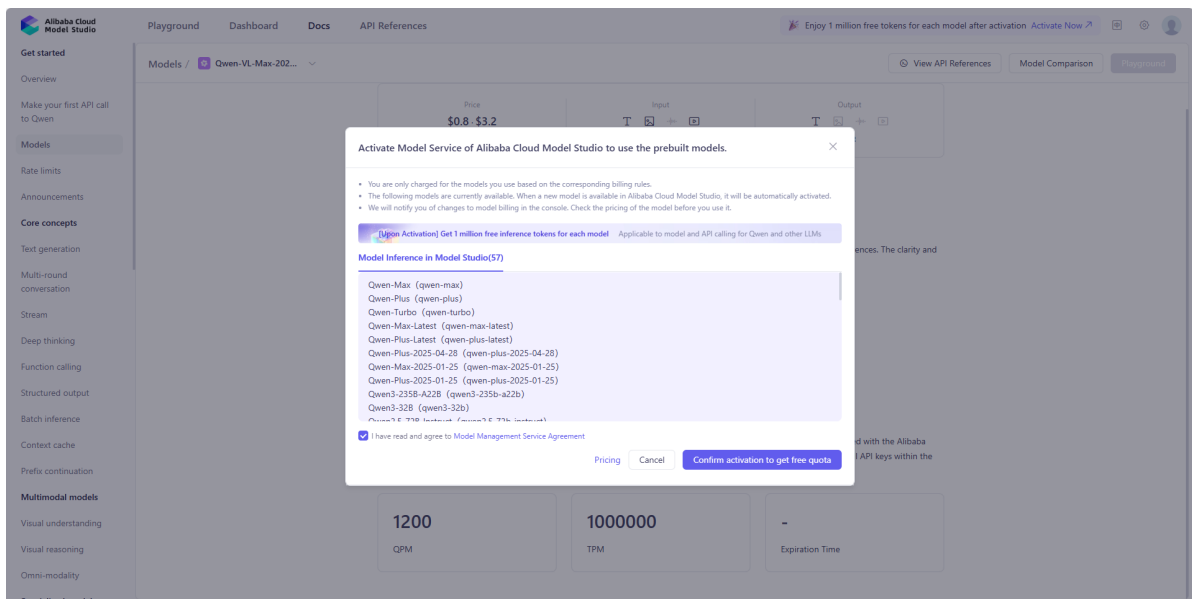
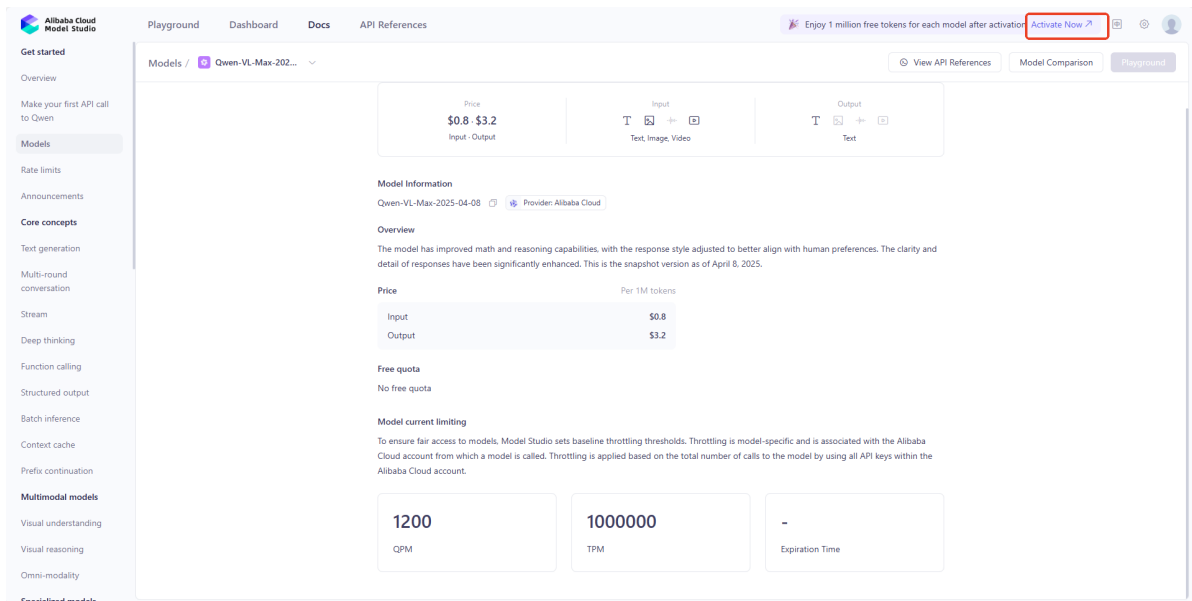
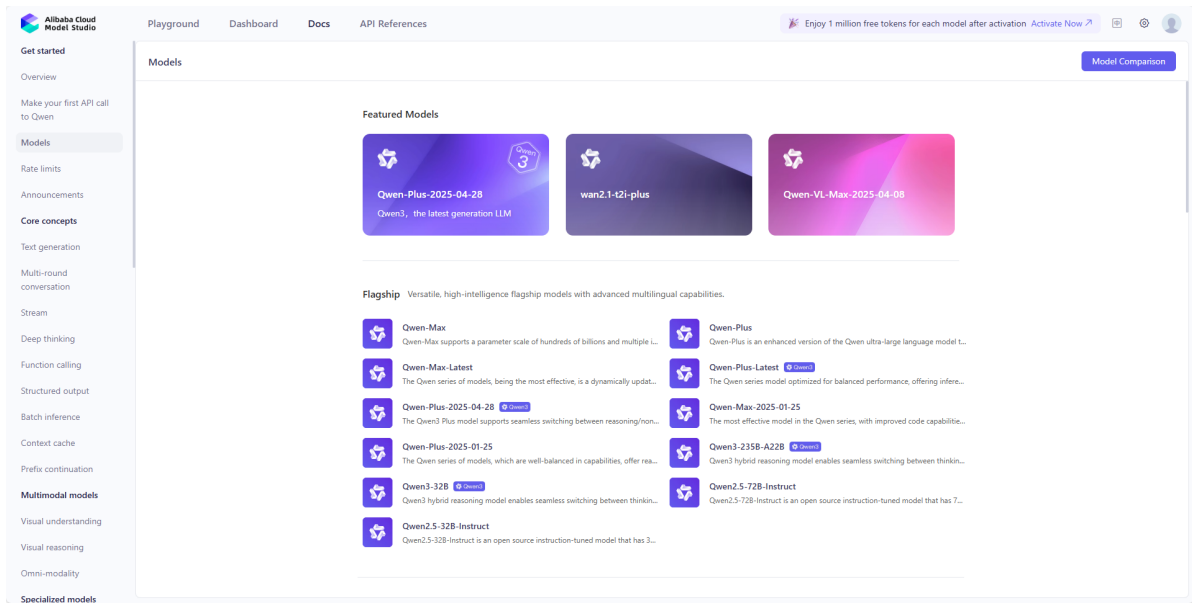
Documentation



Qwen3: Major Models Unveiled!

All Qwen3 models seamlessly integrate "thinking" and "non-thinking" modes, allowing you to switch modes during conversation. [Try It Now in Model Studio!](#)

Chat now with Alibaba Cloud Customer Service to assist you in finding the right products and services to meet your needs.



4. Register your own API Key and OpenAI Compatible Endpoint.

Alibaba Cloud Model Studio

Obtain an API key

Set API key as an environment variable

Install the SDK

Chat

Qwen

Image generation

Wan text-to-image V2

Video generation

Wan image-to-video: first frame

Wan image-to-video: first and last frame

Wan text-to-video

Wan general video editing

Text Embedding

Synchronous API reference

Billing

OpenAI compatibility

Chat

Vision

Playground

Dashboard

Docs

API References

Enjoy 1 million free tokens for each model after activation [Activate Now](#)

Obtain an API key

An API key is needed for authentication when you call large language models (LLMs) or applications through API or SDK. This topic describes how to obtain an API key.

Activate Model Studio

Log on to the [Model Studio console](#). If the following message appears at the top of the page, you need to activate Model Studio and claim the free quota. It is free to activate the service, but after you run out of the quota, you are charged for model calling.

Enjoy 1 million free tokens for each model after activation [Activate Now](#)

Obtain API key

1. Go to [API Key Management](#). Click **Create My API Key**.

2. In the **Actions** column, click **View**. Then, you can view the **API KEY**.

API Key Management

MineAll

Create My API Key (11/10)

ID	API Key	Workspace	Description	Creation Time	Actions
999	sk-2****d1ba	Default Workspace		2024-10-15 10:57:39	ViewDelete

Note

- You can only use the **All API Keys** page if you are using the Alibaba Cloud account. The Alibaba Cloud account can obtain the API keys of all RAM users, but a RAM user can only obtain its own API key.
- Keep your API key confidential to avoid security risks or financial losses caused by unauthorized usage.
- Your API key is an important asset, make sure to keep it safe. If you click **Delete** in the **Actions** column to delete an existing API Key, you cannot use the key to use Model Studio services again. All applications or services associated with this key will fail.
- When creating an API key, we recommend that you select the **default workspace** for **Workspace**. To use API keys of a sub-workspace, you must use the Alibaba Cloud account to [authorize the sub-workspace to use models](#).

Alibaba Cloud Model Studio

Prompts

Permissions

API-Key

默认业务空间

Playground

Dashboard

Docs

API References

Enjoy 1 million free tokens for each model after activation [Activate Now](#)

API Key Management

MyAll

Create My API Key (0/10)

ID	API Key	Workspace	Description	Creation Time	Actions
No data					

Save Your API Key

! You cannot get this key again after you close this dialog box.

Immediately copy or download your API key. Keep this key secure. Anyone with access can make requests on your behalf and incur charges. If the key is lost, reset it or create a new one

API Key

sk-w [redacted] ✓ Copy

API Host

v [redacted] s.com Copy

OpenAI Compatible Endpoint

https:// [redacted] n/compatible-mode/v1 Copy

DashScope

https://w [redacted] n/api/v1 Copy

[Hide](#) ^

Description

-

Workspace

Download Close

Then remotely access the system of the car using VNC and open a terminal. For instructions on how to access the system using VNC, please refer to the tutorial in Chapter 1.

5. Enter in the terminal, Configure API-Key

```
nano /home/pi/project_demo/09.AI_Big_Model/API_KEY.py
```

```

20
21
22 #通义千问 Tongyi Qianwen
23 TONYI_key='sk-b [redacted] :6' #填写通义千问的APIKEY
24
25
26
27 #####国外key相关的 Foreign key related
28 #注册的地址 : https://openrouter.ai/ Registered address : https://openrouter.ai/
29 openAI_KEY = 'sk-or-v1-[redacted] 408d0f' #填写openrouter
   的APIKEY
30

```

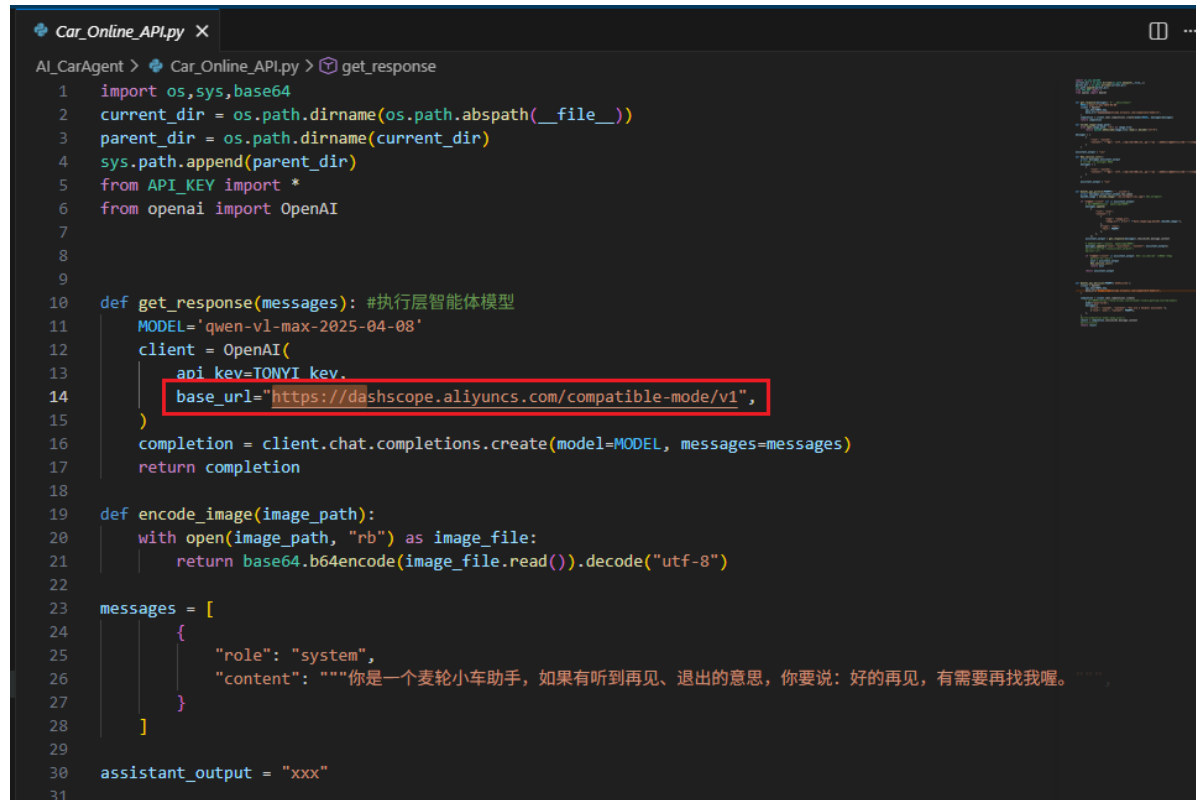
Python 2 ▾ Tab Width: 8 ▾ Ln 6, Col 1 INS

Then, following the prompts, enter the OpenRouter API-KEY and the API-KEY for TONYI_KEY. Press Ctrl+S to save and Ctrl+X to exit.

6. Configure base_url (Tongyi Qwen)

Enter the following command in the terminal to open the file containing the Bailian Platform base_url, then paste the copied OpenAI-compatible address at the base_url field.

```
gedit /home/pi/project_demo/09.AI_Big_Model/AI_CarAgent/Car_Online_API.py
```



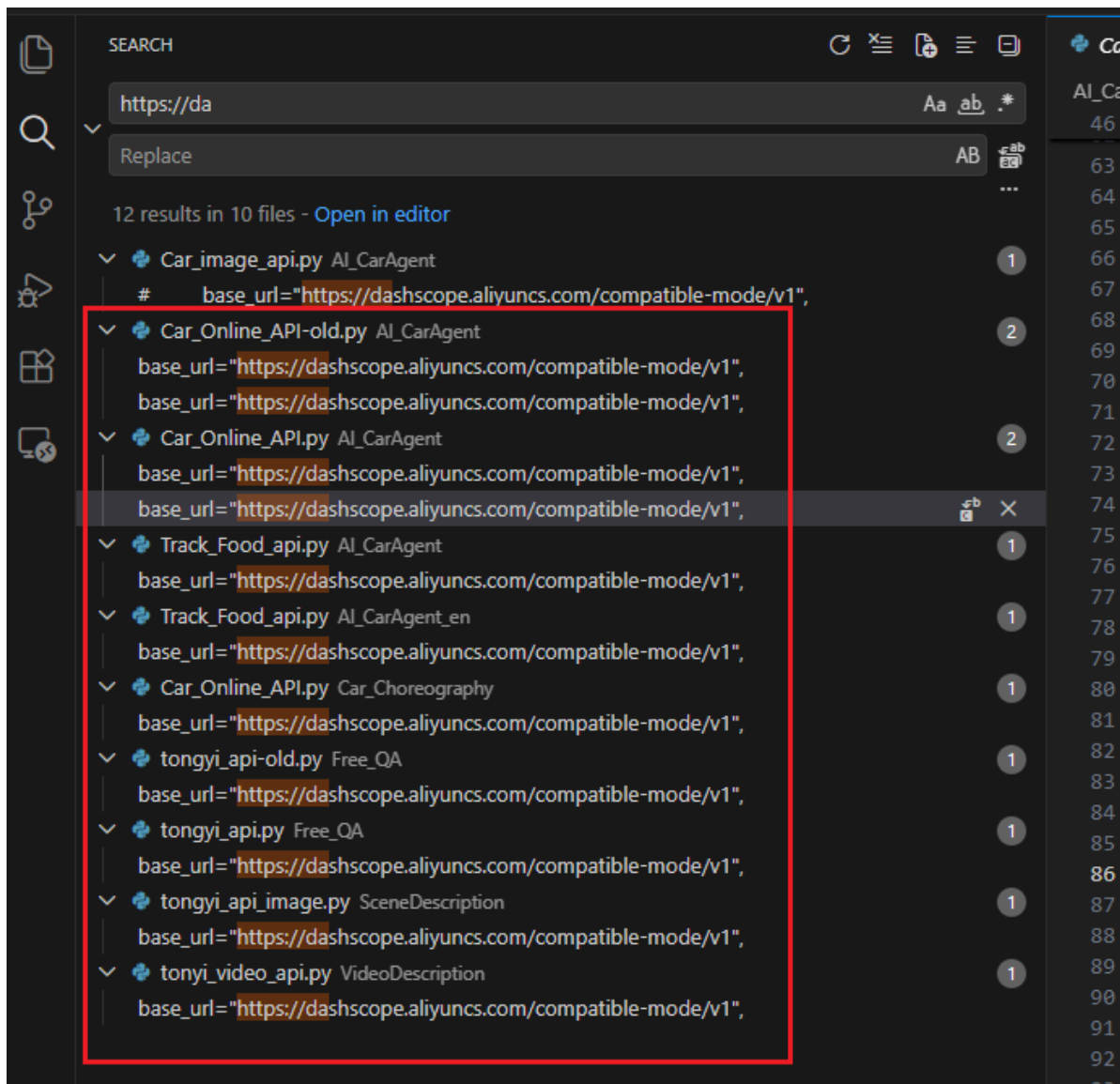
```
Car_Online_API.py X
AI_CarAgent > Car_Online_API.py > get_response
1 import os,sys,base64
2 current_dir = os.path.dirname(os.path.abspath(__file__))
3 parent_dir = os.path.dirname(current_dir)
4 sys.path.append(parent_dir)
5 from API_KEY import *
6 from openai import OpenAI
7
8
9
10 def get_response(messages): #执行层智能体模型
11     MODEL='qwen-vl-max-2025-04-08'
12     client = OpenAI(
13         api_key=TONYI_key,
14         base_url="https://dashscope.aliyuncs.com/compatible-mode/v1",
15     )
16     completion = client.chat.completions.create(model=MODEL, messages=messages)
17     return completion
18
19 def encode_image(image_path):
20     with open(image_path, "rb") as image_file:
21         return base64.b64encode(image_file.read()).decode("utf-8")
22
23 messages = [
24     {
25         "role": "system",
26         "content": """"你是一个麦轮小车助手，如果有听到再见、退出的意思，你要说：好的再见，有需要再找我喔。""",
27     }
28 ]
29
30 assistant_output = "xxx"
31
```

```
AI_CarAgent > Car_Online_API.py > QwenVL_api_decision
46 def QwenVL_api_picture(PROMPT='执行智能体'):
63     ],
64     })
65     assistant_output = get_response(messages).choices[0].message.content
66
67     # 将大模型的回复信息添加到messages列表中
68     messages.append({"role": "assistant", "content": assistant_output})
69     #print(f"模型输出: {assistant_output}")
70     #print("\n")
71
72     if "有需要再找我喔" in assistant_output: #输出结果会在下一个问题前更新
73         #print("清空对话")
74         exit = assistant_output
75         New_session_init()
76         return exit
77
78     return assistant_output
79
80
81
82
83 def QwenVL_api_decision(PROMPT='决策智能体'):
84     client = OpenAI(
85         api_key=TONYI key,
86         base_url="https://dashscope.aliyuncs.com/compatible-mode/v1",
87     )
88
89     completion = client.chat.completions.create(
90         # 模型列表: https://help.aliyun.com/zh/model-studio/getting-started/models
91         model="qwen-turbo",
92         messages=[
93             {"role": "system", "content": "You are a helpful assistant."},
94             {"role": "user", "content": PROMPT},
95         ],
96     )
97     #print(completion.model_dump_json())
98     result = completion.choices[0].message.content
99     #print(result)
100     return result
```

Press Ctrl+S to save and Ctrl+X to exit.

Follow the same procedure to modify other files.

```
gedit /home/pi/project_demo/09.AI_Big_Model/AI_CarAgent/Car_Online_API-old.py
gedit /home/pi/project_demo/09.AI_Big_Model/AI_CarAgent/Car_Food_api.py
gedit /home/pi/project_demo/09.AI_Big_Model/Car_Choreography/Car_Online_API.py
gedit /home/pi/project_demo/09.AI_Big_Model/Free_QA/tongyi_api.py
gedit /home/pi/project_demo/09.AI_Big_Model/Free_QA/tongyi_api-old.py
gedit /home/pi/project_demo/09.AI_Big_Model/SceneDescription/tongyi_api_image.py
gedit /home/pi/project_demo/09.AI_Big_Model/VideoDescription/tonyi_video_api.py
```



You can now run the Big Model demo in the English version.

The Spark Big Model key used by international customers is already included in the relevant examples and does not need to be entered here.

3. Local DIFY Configuration

The image's Dify function is disabled by default. Execute the above command to enable it.

```
cd ~/dify-1.6.0/docker
./docker-compose-linux-aarch64 up -d
```

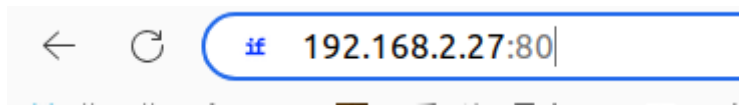
This will enable it.

If you don't want Dify to be running all the time (not recommended for 2GB of RAM), execute the following command to disable it.

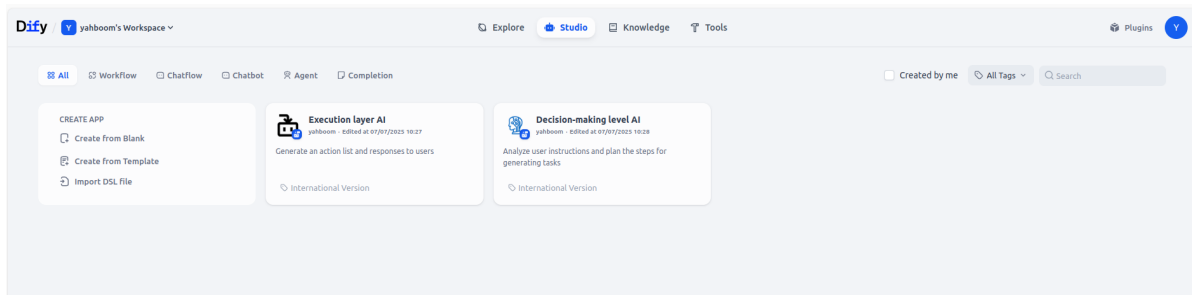
```
cd ~/dify-1.6.0/docker
./docker-compose-linux-aarch64 down
```

Enter the Dify configuration page

- **Observe the IP address of the OLED screen**
- Open a browser on any computer in the Raspberry Pi system on the same network segment as the car and enter the IP address + :80 in the address bar. For example:



The page after entering dify is as follows:

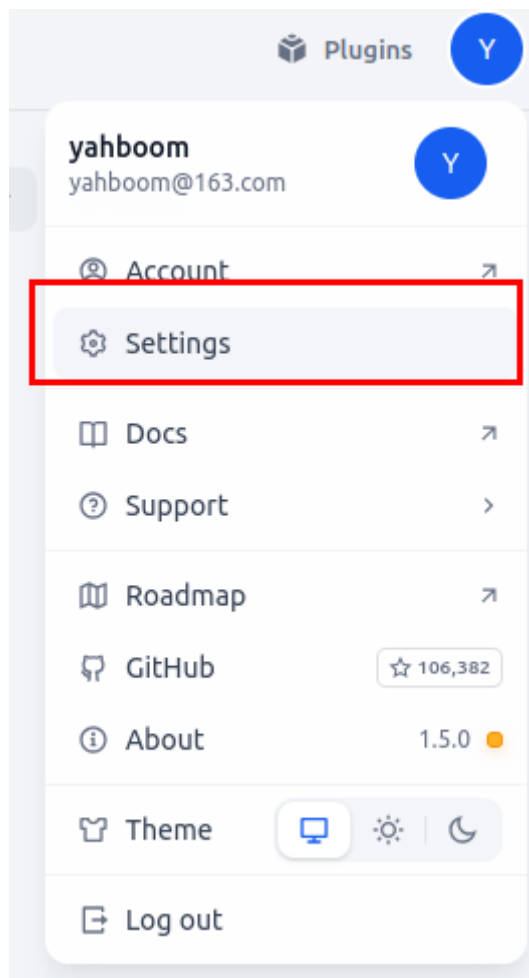


If you are accessing the website from an unfamiliar device, you will need to log in with your account:

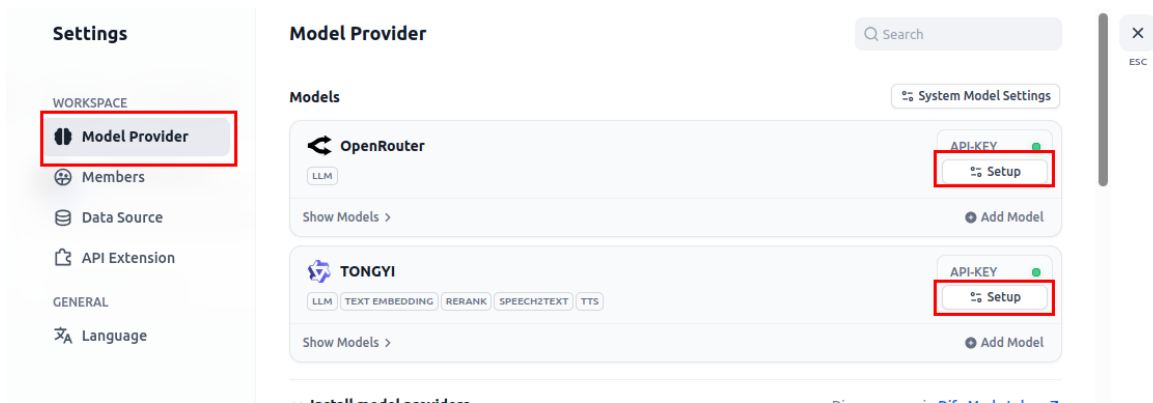
- Username: yahboom@163.com
- Password: yahboom123

3.1 Enter the API key.

Click your profile picture in the upper right corner, then click Settings.



Click Model Provider, then click Setup for the corresponding model provider.



Enter the API key you applied for in [2.3 Registering an OpenRouter Platform Account], then click Save.

Click Setup for TONGYI, then enter the API-KEY you applied for in [2.4 Registering an Alibaba Bailian Model International Platform Account], then select Use International. Endpoint, then click Save.

At this point, set `DIFY_SWITCH=True` in `API_KEY.py` to run the configured agent in dify.

```
API_KEY.py
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30 # 国外 dify 的开关
31 DIFY_SWITCH = True
32
33
34
35
36
37
```